

# A Top-down Graph-based Tool for Modeling Classical Semantic Maps: A Crosslinguistic Case Study of Supplementary Adverbs

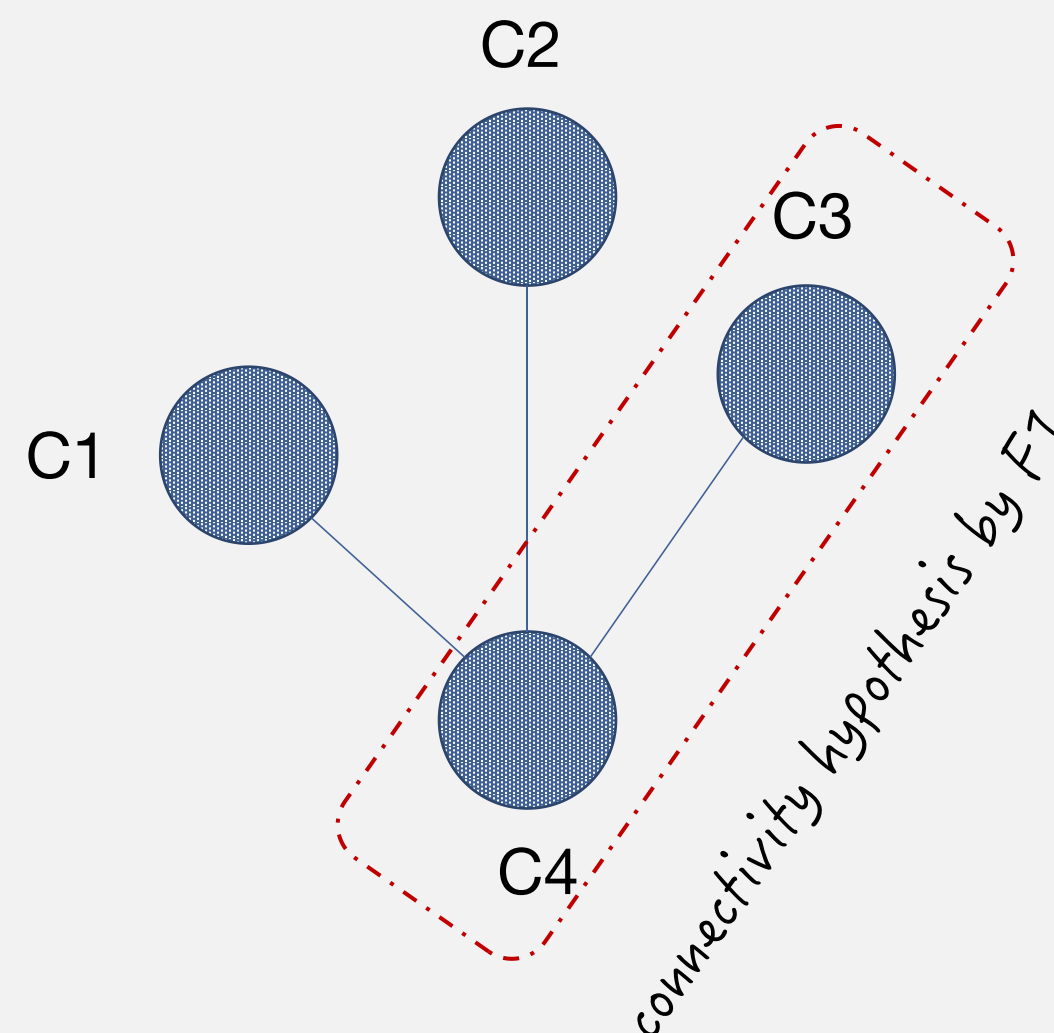
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## What are classical semantic map models?



|    | C1 | C2 | C3 | C4 |
|----|----|----|----|----|
| F1 |    |    | ✓  | ✓  |
| F2 |    | ✓  |    | ✓  |
| F3 | ✓  |    |    | ✓  |
| F4 | ✓  |    |    | ✓  |

Form-Concept/Function Table



Semantic Map Models

**Node:** Concept / Meaning / Function / Sense

**Forms:** Function words or content words / Affixes / Constructions

**Edge:** Association between nodes

**Form-Concept Table:** Patterns of multifunctionality/Polysemous

**Connectivity Hypothesis (H1):** concepts associated with a single form must be connected!

**Coverage vs. Predictivity Dilemma**

**Vacuous Map:** A cyclic graph that makes no predictions - avoid it!

Croft, William 2003 Typology and Universals(2nd edition). Cambridge: Cambridge University Press.

**Domain: External-possession construction**

**English preposition to**

- Goethe went to Vienna as a student. (direction)
- Eve gave the apple to Adam. (recipient)
- This seems outrageous to me. (experiencer)
- I left the party early to get home in time. (purpose)

**English preposition for**

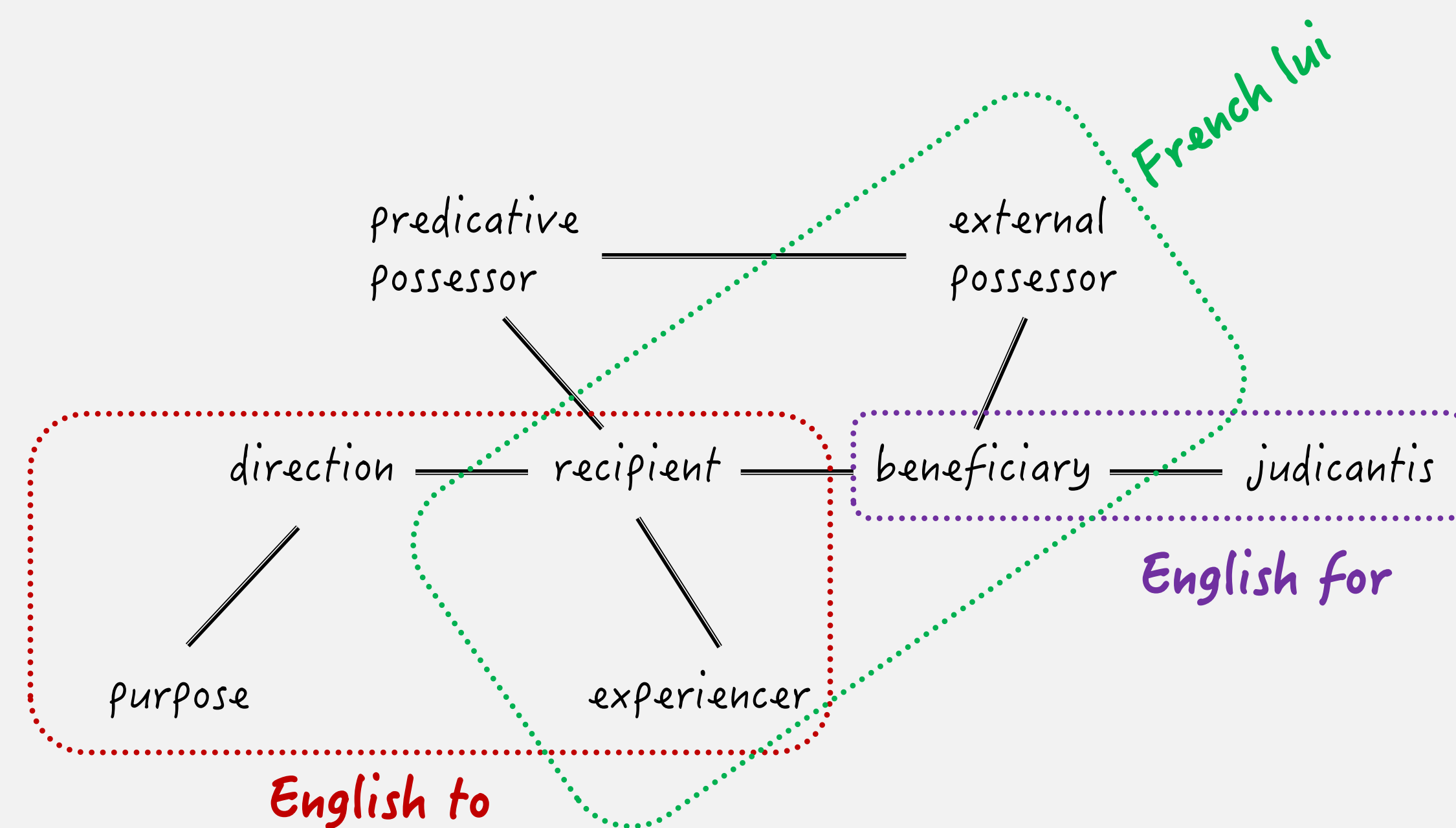
- My mom bought a bike for the son. (benefactive)
- The math is too difficult for me. (judicantis)

**French lui (DAT-case)**

- Je lui donne le livre. [I'll give her the book] (recipient)
- Je lui ai trouvé unemploi [I found a job for him.] (benefactive)
- On lui a cassé la jambe [They broke his leg] (ext. possessor)
- Ce livre lui plaît [He likes this book] (experiencer)

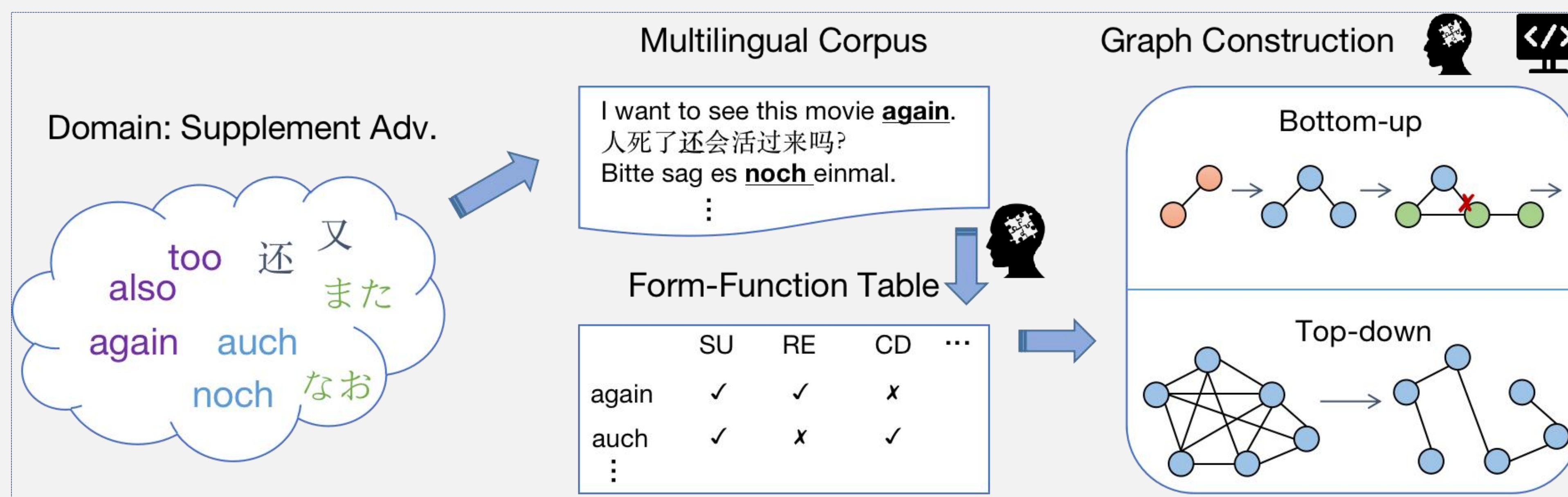
Russian: u, DATive-case

Chinese, Catalan, Homeric Greek, Basque ...



Haspelmath M. External possession in a European areal perspective[J]. Typological studies in language, 1999, 39: 109-136.

## How to construct classical semantic maps?



- Bottom-up approach
- Expert knowledge or prior programs
  - Incrementally satisfy connectivity hypothesis (H1) case by case
  - Avoid generating cycles (vacuous graphs) when necessary
  - Time-consuming and hard to scale ✗

- Top-down approach (ours)
- Initially generate a complete graph by calculating the number of bifunctional co-occurrences
  - Convert local connectivity H1 into a global one H2
    - Efficient and automated ✓
- H2: A Maximum spanning tree from the complete graph
- Overall connectivity
  - Acyclicity
  - Maximum total edge weights

## Evaluations on a case study

### Overview

- 9 languages, 28 grammatical forms and 18 functions
- Supplementary Adverbs (The core meaning is to express supplementation)
- The form-function table has been created by linguists in a bottom-up fashion.
- The golden map (GT) serves as a reference standard

### Metrics

- Size: the summed of edge weights
- Recall: The proportion of cases satisfying H1 out of all cases
- Precision: The proportion of cases satisfying H1 among all possible cases generated by the graph.
- Accuracy: The proportion of matches compared to the reference.
- Standard deviation of degrees: prefers a chain-like typology

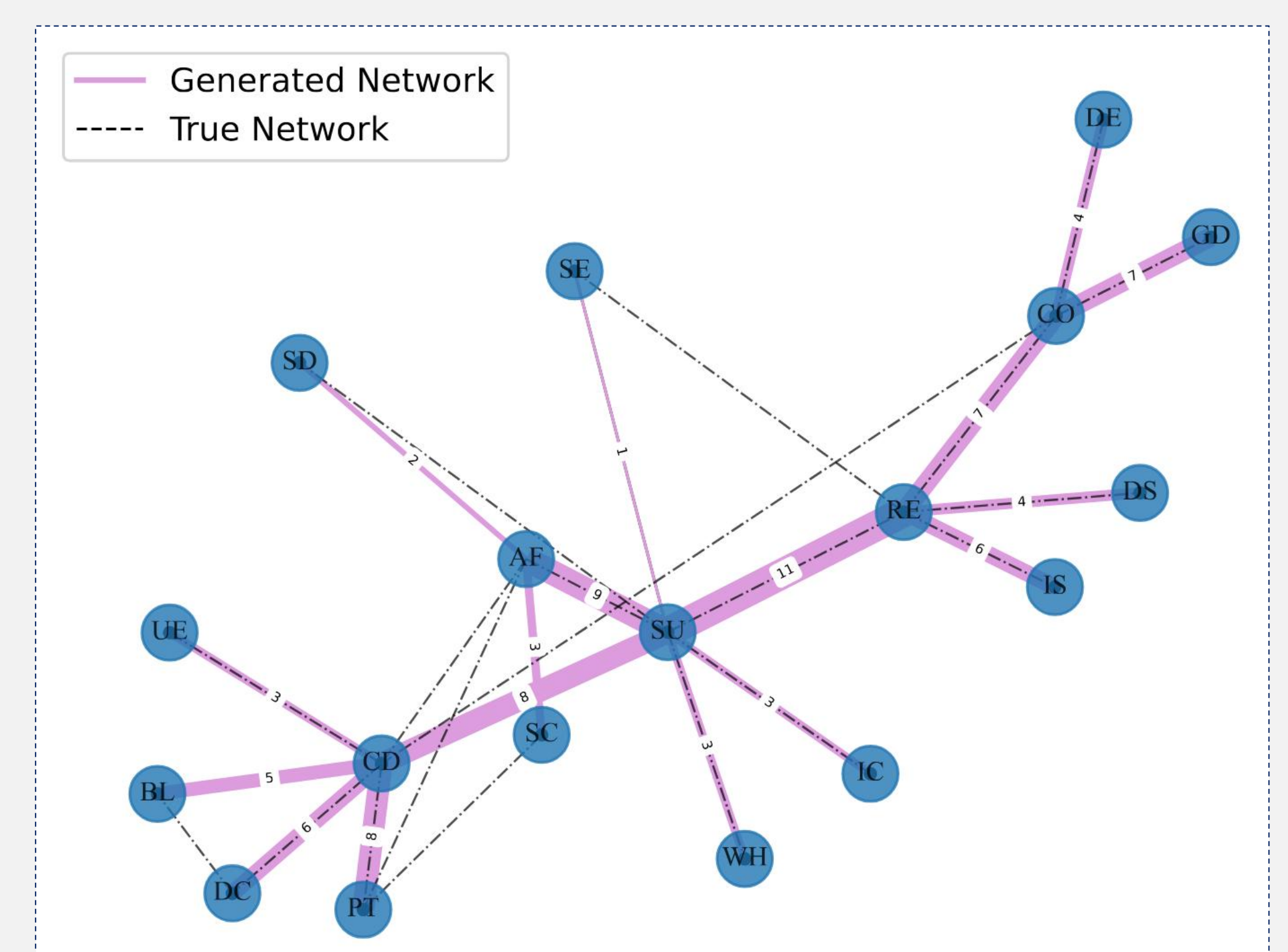
### Baselines

- C: A complete graph
- LT: A lower-bound baseline with a 'bad' tree that does not overlap with GT.

| Index | Size↑ | Recall↑ | Precision↑ | Accuracy↑ |
|-------|-------|---------|------------|-----------|
| C     | 286   | 1       | 0          | 50.0      |
| LT    | -     | -       | -          | 79.0      |
| GT    | 91    | 1       | 0.20       | 1         |
| 0     | 90    | 85.7    | 0.17       | 92.6      |
| 1     | 89    | 82.1    | 0.21       | 91.4      |
| 2     | 89    | 82.1    | 0.44       | 90.1      |
| 3     | 88    | 82.1    | 0.34       | 91.4      |
| 4     | 88    | 78.6    | 0.50       | 88.9      |

| Round     | RG_1  | RG_2  |
|-----------|-------|-------|
| 1         | -17.8 | -22.1 |
| 2         | -21.9 | -22.4 |
| 3         | -20.5 | -19.2 |
| 4         | -23.8 | -21.7 |
| 5         | -23.1 | -24.1 |
| Mean      | -21.4 | -21.9 |
| Std. Dev. | 2.13  | 1.58  |

- A graph similar to expert-generated ones, achieving high accuracy.
- Multiple candidate solutions
- A metric with a moderate correlation to quality.
- Edge failures are primarily due to the acyclicity constraint (e.g., CD).



Rui Guo. 2010. A semantic map study of adverbs related to "supplement". In Paper presented at the International Symposium for Comparative and Typological Research on Languages of China, Hong Kong, China.

